MODULE 08 · DOMAIN 1 - PLAN AND MANAGE AN AZURE AI SOLUTION

Monitoring models and retrieval

Platform metrics, safety events, continuous and scheduled evaluation, drift, grounding quality, and search index health.

DOMAIN 1 - Plan and manage WEIGHT 25-30% MODULE 08 of 39

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Healthy dashboards can hide bad answers. Monitor **the platform**, **the answers**, and **the retrieval** behind them.

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Three layers, three sets of signals

LAYER	QUESTION IT ANSWERS	SIGNALS
Platform	Is the service fast, available and within capacity?	Azure Monitor metrics and resource logs
Answers	Are responses good, grounded and safe?	Continuous and scheduled evaluation, safety metrics
Retrieval	Is the right content indexed and found?	Search metrics, indexer history, retrieval evaluators

ON THE EXAM

When a question names a symptom, first decide which layer it lives in. That rules out most of the options.

Azure Monitor basics

- **Platform metrics** and the activity log are collected automatically.
- **Resource logs are not collected** until you create a **diagnostic setting** to route them.
- Route logs to a Log Analytics workspace, a storage account, or Event Hubs.
- Split metrics by **ModelDeploymentName** to see each deployment separately.
- Alert rules send notifications through **action groups**: email, SMS, or an incident tool.

> ON THE EXAM
>
> 'We need the logs from last Tuesday, but no diagnostic setting existed' has no fix. Logs cannot be collected retrospectively.

The metrics that matter for models

TO MONITOR	USE
Overall response time	Time to Last Byte
First-token responsiveness, when streaming	Time to Response
Generation speed	Time Between Tokens
Input and output volume	Processed Prompt Tokens, Generated Completion Tokens
Provisioned capacity use	Provisioned-managed Utilization V2
Request volume and errors	Azure OpenAI Requests, split by status code

WATCH FOR THIS

The legacy 'Latency' metric under Cognitive Services HTTP Requests is misleading for these workloads. Don't diagnose with it.

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Latency up **with** tokens up is expected.
Latency up with tokens **flat** is a real problem.

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Watching the safety system

- **Blocked Volume:** requests or responses blocked by content filters.
- **Harmful Volume Detected:** content flagged as harmful.
- **Potential Abusive User Count:** users showing a pattern of abuse.
- **Safety System Event** and **Total Volume Sent for Safety Check** give context and scale.
- Alert on a **rising rate**, not a single block. One blocked request is the system working.

> ON THE EXAM
>
> A sudden rise in blocked volume after a release may mean users are attacking, or that a prompt change is tripping filters.

Continuous or scheduled evaluation?

CONTINUOUS EVALUATION

Real traffic, sampled

- Scores a sample of live production traffic
- Catches problems real users actually hit
- Scores move as the mix of questions changes
- Default limit of 100 runs per hour

SCHEDULED EVALUATION

Fixed dataset, repeated

- Runs the same test dataset on a schedule
- Same inputs every time, so scores are comparable
- Microsoft's stated tool for detecting drift
- Catches slow decline nobody reported

SETUP DETAIL

The project's managed identity needs a data plane role, such as Foundry User, to run evaluation rules.

Drift means **the same questions** get worse answers over time. Only a fixed dataset, rerun on a schedule, can prove it.

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Foundry's production observability

COMPONENT	WHAT IT ADDS
Operational metrics	Tokens, latency, success rates and errors
Continuous evaluation	Quality and safety on sampled live traffic
Scheduled evaluation	Fixed-dataset scores over time, to detect drift
Scheduled red teaming	Regular adversarial probing for new weaknesses
Azure Monitor alerts	Notification when quality or safety crosses a threshold

ON THE EXAM

The Agent Monitoring Dashboard needs Application Insights connected to the project. Agents hosted elsewhere can send OpenTelemetry to the same instance.

Evaluators for RAG quality

EVALUATOR	ASKS	GROUND TRUTH?
Retrieval	Were retrieved chunks relevant?	No
Groundedness	Does the answer stay within context?	No
Relevance	Does the answer address the question?	No
Response Completeness (preview)	Did it miss anything important?	Yes
Document Retrieval	How good was the ranking?	Yes, labels

ON THE EXAM

Microsoft's recommended set for RAG apps: Retrieval, Groundedness, Relevance, plus Content Safety.

Reading groundedness and relevance together

GROUNDED, BUT NOT RELEVANT

Faithful to the wrong documents

- Answer sticks closely to retrieved text
- But the text doesn't answer the question
- Look at retrieval: index, freshness, query

RELEVANT, BUT NOT GROUNDED

On topic, but invented

- Answer addresses the question well
- But claims aren't in the retrieved context
- Look at the model: instructions, fabrication

THE TIEBREAKER

This pairing is a diagnostic habit, not a formula. It tells you where to look first.

Azure AI Search health signals

METRIC	WHAT IT TELLS YOU
Search Latency	How long queries take to complete
Search Queries Per Second	Query volume, per one-minute window
Throttled Search Queries Percentage	Share of queries dropped rather than processed
Storage usage, by index	Growth towards the service tier's limits
Documents processed, skill executions	Indexing progress and enrichment activity

ON THE EXAM

Some throttling is normal, especially during indexing, when queries and indexing share the same resources.

Is the right content actually indexed?

- **Indexer execution history** shows the last 50 runs: status, times, errors and warnings.
- **Document-level errors** name the files that failed, and why. Export them as CSV.
- Warnings are often harmless, such as an image or PDF with no text to extract.
- Compare the **indexed document count** with the source. A gap means content is missing.
- Fast, healthy queries over an incomplete index still produce wrong answers.

> THE SIGNAL
>
> Search latency and throttling alerts say nothing about whether new documents were indexed.

Tracing a bad answer, layer by layer

01
Platform
Errors, throttling, or latency without more tokens?

›

02
Ingestion
Did the indexer run cleanly, with full counts?

›

03
Retrieval
Were relevant chunks retrieved?

›

04
Grounding
Did the answer stay within them?

›

05
Relevance
Did it answer the question asked?

ON THE EXAM

Work from the bottom of the stack upwards. A retrieval failure makes every layer above it look broken.

> After a prompt change asking for more detailed answers, an operations dashboard shows the legacy Latency metric rising sharply. The team suspects the model is degraded.
>
> Azure OpenAI metrics show Time to Last Byte has risen by about the same proportion as Generated Completion Tokens, which doubled.

What should the team conclude?

A. It's expected, as outputs are longer; chart latency with tokens.

B. The deployment is saturated, so it should move to provisioned throughput.

C. The service is degraded, so the team should open a support request.

D. The region is overloaded, so the deployment should move to another region.

◆ PAUSE AND DECIDE

What should the team conclude?

A Expected; pair latency with tokens.

Response time rose in proportion to output length, which Microsoft describes as expected behaviour.

B Saturated; move to provisioned.

Nothing shows throttling or capacity pressure. The extra time is extra tokens.

C Degraded; open a support case.

The evidence points to the team's own prompt change, not a service fault.

D Overloaded region; move it.

Moving regions wouldn't shorten the outputs that are taking longer to generate.

THE SIGNAL

Two clues: the legacy metric is unreliable, and latency moved with token count. Together they say 'expected'.

> Users say a support agent's answers have become worse over two months. Continuous evaluation scores fluctuate, but the mix of customer questions also shifted during a product launch.
>
> Leadership wants evidence of whether the agent itself has declined.

What should the team do?

A Raise continuous evaluation sampling so every production request is scored.

B Review the trend in processed prompt and generated completion tokens.

C Run a scheduled evaluation on a fixed dataset and compare over time.

D Create a diagnostic setting to send request logs to Log Analytics.

◆ PAUSE AND DECIDE

What should the team do?

A Score every production request.

More samples of a changing question mix. Still can't separate harder questions from a worse agent.

B Review token trends.

A platform signal. Token counts say nothing about answer quality.

C Scheduled evaluation on a fixed dataset.

Same questions over time, so a lower score means the agent changed. Microsoft's drift tool.

D Diagnostic setting for request logs.

Useful for operations, but logs don't score quality, and can't cover the past two months.

THE SIGNAL

'The question mix also shifted' is the planted detail. It makes live-traffic scores ambiguous, and a fixed dataset the answer.

> A legal team adds new policy PDFs to Blob Storage each week. Its agent answers old policy questions well, but cannot answer anything about the new policies.
>
> The team alerts on search latency and throttled queries, and both look healthy. The indexer history shows document-level errors on recent runs.

What should the team add to its monitoring?

A. A tighter alert threshold on search latency for the policy index.

B. More search replicas, to reduce throttled queries while indexing runs.

C. A higher reasoning effort setting on the agent's model deployment.

D. Alert on indexer errors and compare document counts with the source.

◆ PAUSE AND DECIDE

What should the team add to its monitoring?

A Tighter latency threshold.

Latency is healthy. Queries run quickly over an index that is missing documents.

B More replicas.

Throttling isn't the problem, and replicas don't fix failed documents.

C Higher reasoning effort.

No amount of reasoning finds content that was never indexed.

D Indexer error alerts and count checks.

Watches ingestion quality directly, catching failed documents before users do.

THE SIGNAL

Old content works, new content fails: that split points at ingestion, which query metrics cannot see.

> A sales assistant's evaluation run scores high on groundedness but low on relevance for questions about 2026 pricing.
>
> Reviewers see that answers accurately quote company documents, but the documents quoted describe 2024 pricing.

Where should the team investigate first?

A. The model, which is fabricating content; switch to a larger model.

B. Retrieval, which returns old documents; check index freshness.

C. The content filters, which may be blocking current pricing content.

D. The groundedness evaluator, which is scoring these answers incorrectly.

◆ PAUSE AND DECIDE

Where should the team investigate first?

A. Fabrication; larger model.
High groundedness means the model is not inventing. It is faithfully quoting what it was given.

B. Retrieval returning outdated documents.
Grounded but irrelevant: the model used the context well, but the context was wrong.

C. Content filters.
Pricing isn't harmful content, and blocked content would show in safety metrics.

D. A faulty evaluator.
Reviewers confirmed the quotes are accurate. The evaluator is right.

THE TIEBREAKER

Grounded but irrelevant is the classic signature of a retrieval problem. Fix what the model is given, not the model.

Monitoring in five lines

- **Three layers:** platform, answers, retrieval. Healthy metrics can hide bad answers.
- **Resource logs need a diagnostic setting.** Pair latency with token counts.
- **Continuous evaluation** samples live traffic. **Scheduled evaluation** on fixed data detects drift.
- **Grounded but irrelevant** points to retrieval. Relevant but ungrounded points to the model.
- **Watch ingestion**: indexer errors and document counts, not just query speed.

> ON THE EXAM
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> Next, module 09: configuring guardrails, content filters, blocklists and prompt shields.